1.3 The Limit of a Function

Calculus 1

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Textbook: James Stewart, Essential Calculus - Early Transcendentals

Outline

1. Intuitive Definition of a Limit
2. One-sided Limits
3. Precise Definition of a Limit

< Part 1 >

Intuitive Definition of a Limit

Intuitive Definition of a Limit

In general, we use the following notation.

Definition

Suppose $f(x)$ is defined when x is near the number a. Then

$$\lim_{x \to a} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a but not equal to a.

Intuitive Definition of a Limit

The values of $f(x)$tend to get closer and closer to the number L as x gets closer and closer to a (from either side of a) but $x \neq a$.

Notice the phrase "but $x \neq a$" in the definition of limit.

This means that we never consider $x = a$.

The only thing that matters is how f is defined near a.

Example 2

Guess the value of $\lim_{x\to 1} \frac{x-1}{x^2-1}$.

Solution:

Notice that the function $f(x) = (x-1)/(x^2-1)$ is not defined when $x = 1$, but that doesn't matter because ...

the definition of $\lim_{x\to a} f(x)$ consider values of x that are close to a but not equal to a.

Example 2 - Solution

Values of $f(x)$ for values of x that approach 1 (but are not equal to 1).

$x < 1$	$f(x)$	$x > 1$	$f(x)$
0.5	0.666667	1.5	0.400000
0.9	0.526316	1.1	0.476190
0.99	0.502513	1.01	0.497512
0.999	0.500250	1.001	0.499750
0.9999	0.500025	1.0001	0.499975

We guess $\displaystyle \lim_{x\to 1} \frac{x-1}{x^2-1} = 0.5$

Example 4

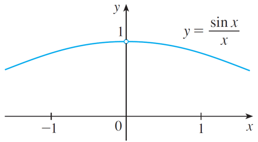
Guess the value of $\lim_{x \to 0} \frac{\sin x}{x}$.

Solution:

The function $f(x) = (\sin x)/x$ is not defined when $x = 0$.

Example 4 - Solution

x	$\sin x/x$	x	$\sin x/x$
± 1.0	0.84147098	± 0.1	0.99833417
± 0.5	0.95885108	± 0.05	0.99958339
± 0.4	0.97354586	± 0.01	0.99998333
± 0.3	0.98506736	± 0.005	0.99999583
± 0.2	0.99334665	± 0.001	0.99999983



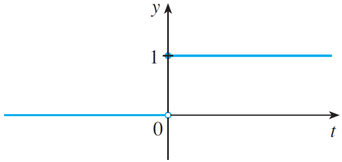
From the table and the graph above we guess that

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

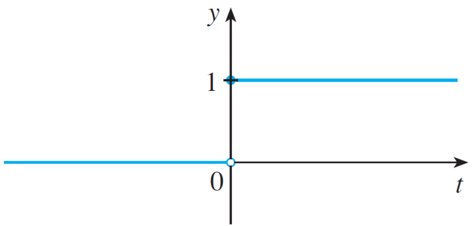
Example 6

The Heaviside function H is defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geqslant 0 \end{cases}$$



Example 6



As $t$approaches 0 from the left, $H(t)$ approaches 0.

As$t$approaches 0 from the right, $H(t)$approaches 1.

There is no single number that $H(t)$ approaches as t approaches 0.

Therefore, $\lim_{t \to 0} H(t)$ does not exist.

Note

< Part 2 >

One-sided Limits

One-sided Limits

Recall that $H(t)$ approaches 0 as t approaches 0 from the left and $H(t)$ approaches 1 as t approaches 0from the right.

Notation:

$$\lim_{t \to 0^-} H(t) = 0, \qquad \lim_{t \to 0^+} H(t) = 1.$$

"$t \to 0^-$" indicates that we consider only values of t that are less than 0.

"$t \to 0^+$" indicates that we consider only values of t that are greater than 0.

One-sided Limits

Definition

We write

$$\lim_{x \to a^-} f(x) = L$$

and say the left-hand limit of $f(x)$ as x approaches a is equal to L if $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a with $x < a$.

Notice that Definition 2 differs from Definition 1 only in that we require x to be less than a.

One-sided Limits

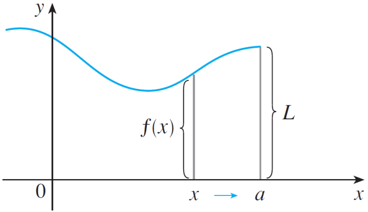
Similarly, if we require that x be greater than a, we get "the **right-hand limit of** $f(x)$ **as** x **approaches** a is equal to L"

Notation:

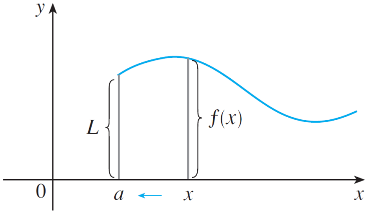
$$\lim_{x \to a^+} f(x) = L.$$

"$x \to a^+$" means that we consider only $x > a$.

One-sided Limits



(a) $\lim_{x \to a^-} f(x) = L$



(b) $\lim_{x \to a^+} f(x) = L$

$$\lim_{x \to a^-} f(x) = L, \qquad \lim_{x \to a^+} f(x) = L$$

Limit vs One-sided Limits

Theorem

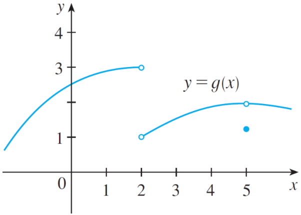
$$\lim_{x \to a} f(x) = L \quad \text{if and only if} \quad \lim_{x \to a^-} f(x) = L \quad \text{and} \quad \lim_{x \to a^+} f(x) = L.$$

Example 7

The graph of a function g is shown in Figure 10. Use it to state the values (if they exist) of the following:

(a) $\lim_{x\to 2^-} g(x)$ **(b)** $\lim_{x\to 2^+} g(x)$ **(c)** $\lim_{x\to 2} g(x)$

(d) $\lim_{x\to 5^-} g(x)$ **(e)** $\lim_{x\to 5^+} g(x)$ **(f)** $\lim_{x\to 5} g(x)$



Example 7 - Solution

(a) $\lim_{x\to 2^-} g(x) = 3$

(b) $\lim_{x\to 2^+} g(x) = 1$

(c) Since the left and right limits are different, we conclude that $\lim_{x\to 2} g(x)$ does not exist.

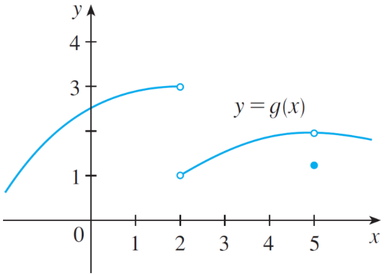
Example 7 - Solution

(d) $\lim_{x \to 5^-} g(x) = 2$

(e) $\lim_{x \to 5^+} g(x) = 2$

(f) This time the left and right limits are the same and so $\lim_{x \to 5} g(x) = 2$

Despite this fact, notice that $g(5) \neq 2$.



Note

< Part 3 >

Precise Definition of a Limit

Precise Definition of a Limit

We want to express that $f(x)$ can be made arbitrarily close to L by taking x to be sufficiently close to a (but $x \neq a$).

This means that $f(x)$ can be made to lie within any preassigned distance from L (traditionally denoted by ϵ, the Greek letter epsilon) by requiring that x be within a specified distance δ (the Greek letter delta) from a.

That is, $(f(x) - L) < \epsilon when (x - a) < \delta and x \neq a$.

Notice that we can use $0 < (x - a) < \delta$.

Precise Definition of a Limit

Definition

Let f be a function defined on some open interval that contains a, except possibly at a itself. We say that the **limit of** $f(x)$ **as** x **approaches** a **is** L, and we write

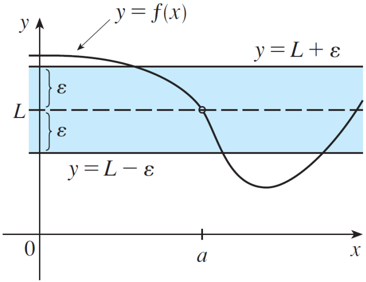
$$\lim_{x \to a} f(x) = L$$

if for every number $\epsilon > 0$ there is a number $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \epsilon.$$

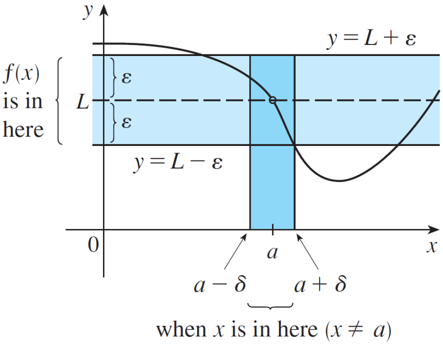
Figure

If a number $\epsilon > 0$ is given, then we draw the horizontal lines $y = L + \epsilon$ and $y = L - \epsilon$ and the graph of f.



Figure

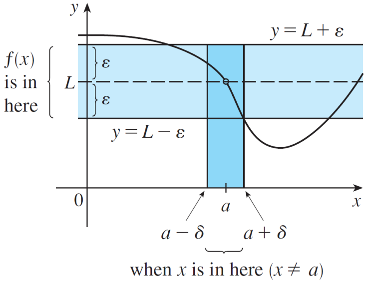
If $\lim_{x\to a} f(x) = L$, then we can find a number $\delta > 0$ such that if we restrict x to lie in the interval $(a-\delta, a+\delta)$ and take $x \neq a$, then the curve $y = f(x)$ lies between the lines $y = L - \epsilon$ and $y = L + \epsilon$.

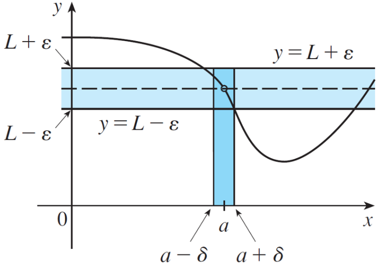


Figure

The process illustrated in the below Figure must work for every positive number ϵ, no matter how small it is chosen.

If a smaller ϵ is chosen, then a smaller δ may be required.





Example 9

Prove that $\lim_{x\to 3}(4x-5)=7$.

Solution:

Let ϵ be a given positive number.

We need to find a number δ such that

$$\text{if } 0 \leq |x-3| < \delta, \quad \text{then} |(4x-5)-7| < \epsilon.$$

Note that

$$|(4x-5)-7| = 4|x-3| < 4\delta =: \epsilon$$

So we can choose $\delta = \epsilon/4$. DONE!

Exercise

Prove that $\lim_{x\to 3} x^2 = 9$.

Solution:

Note